

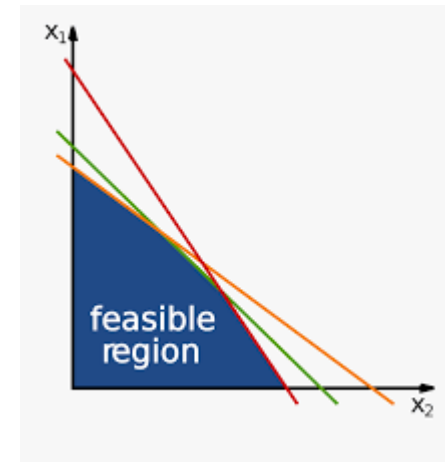


NTNU – Trondheim
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TET4185, Power Markets, Resources and Environment

Introduction to Linear Programming

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Literature recommendation

📖 An excellent (and extensive) book on linear programming is

– Vanderbei, R. J. (2008).

Linear Programming-Foundations and Extensions (3rd ed.).

New York: Springer.

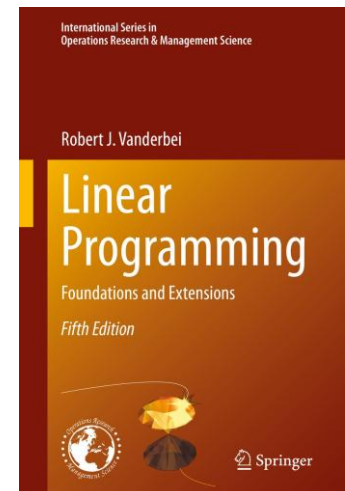
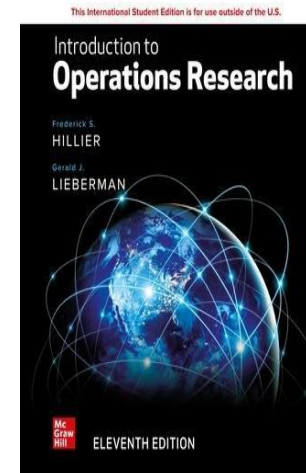
(<https://link.springer.com/book/10.1007/978-3-030-39415-8>)

📖 A book with a more general OR scope is

– Hiller, F. S. & Lieberman G. J. (2010).

Introduction to Operations Research (11th ed.).

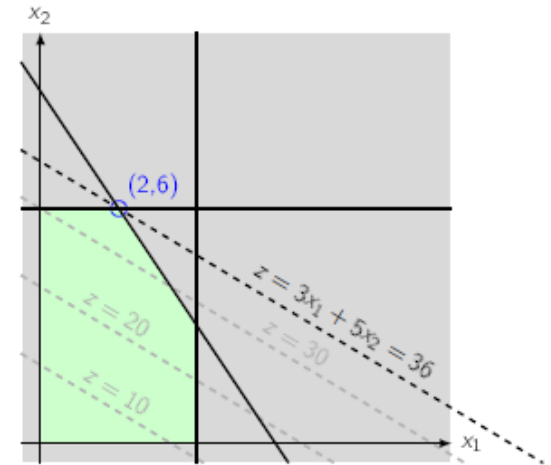
Boston: McGraw-Hill.



Fixing ideas - What is optimization and how does LP fit into this realm?

Optimization for us can be described as:

- given a set of possible decisions,
 - constraints
 - feasible (or admissible) decisions
- and a measure of how good these decisions are,
 - objective function
- find the optimal decisions
 - optimal value of objective function
 - what are the decisions that provide this optimal value



Fixing ideas -What is optimization and how does LP fit into this realm?

- Mathematically we formulate an optimization problem as

Generic (constrained) optimization problem

minimize/maximize $f(x)$, $x \in \mathbb{R}^n$

Subject to

$$\begin{array}{ll} h_i(x)=0, & i=1, \dots, m_1 \\ g_i(x) \leq 0, & i=m_1+1, \dots, m \end{array}$$

Fixing ideas - What is optimization and how does LP fit into this realm?

In a linear program everything is, well, linear.

Generic (constrained) LP optimization problem

minimize/maximize $f(x) = \sum_{j=1}^n c_j x_j$, $x \in \mathbb{R}^n$

Subject to

$$\begin{aligned} h_i(x) &= \sum_{j=1}^n a_{ij} x_j - b_j = 0, & i &= 1, \dots, m_1 \\ g_i(x) &= \sum_{j=1}^n a_{ij} x_j - b_j \leq 0, & i &= m_1 + 1, \dots, m \end{aligned}$$

Linear programs in standard form using matrix notation

Summation form

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

Matrix form

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0. \end{aligned}$$

Where

$$\begin{aligned} \mathbf{x} &= (x_1, \dots, x_n) \\ \mathbf{c} &= (c_1, \dots, c_n) \\ \mathbf{b} &= (b_1, \dots, b_m)^T \\ \mathbf{A} &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \end{aligned}$$

Resource allocation and production example

Door producing company (Hiller & Lieberman 2010)

A company makes two types of doors,

- ① Aluminium framed glass doors
 - ✓ Selling price: \$3000 a piece
 - ✓ Frame takes one hour to make
 - ✓ Assembly of frame and glass takes three hours
- ② Wood framed glass doors
 - ✓ Selling price: \$5000 a piece
 - ✓ Frame takes two hours to make
 - ✓ Assembly of frame and glass also takes two hours

The company rents production capacity at three different factories

- ☐ An aluminium frame production facility, available capacity: 4 h/week
- ☐ A wood frame production facility, available capacity: 12 h/week
- ☐ A frame and glass assembly facility, available capacity: 18 h/week

How can the company decide on how many doors of each type to produce every week?

Resource allocation and production example

Easy! They formulate this as a linear program.

Door producing company (Hiller & Lieberman 2010)

□ Decision variables

1. x_1 : # of aluminium framed glass doors
2. x_2 : # of wood framed glass doors

□ Objective function (maximize revenues):

$$\text{maximize } z = 3000 x_1 + 5000 x_2$$

□ Constraints:

1. Aluminium frame factory capacity:

$$x_1 \leq 4$$

2. Wood frame factory capacity:

$$2x_2 \leq 12$$

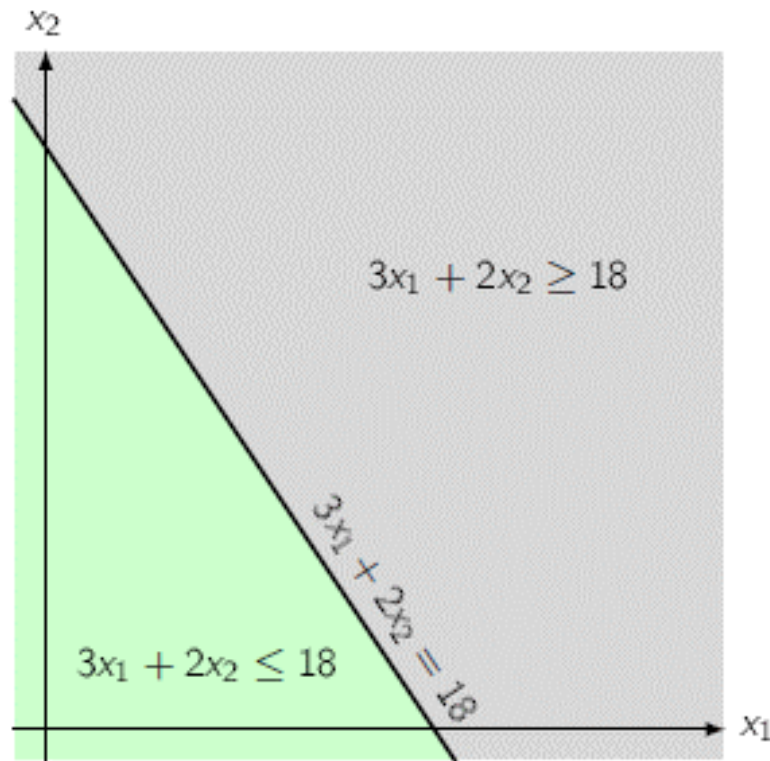
3. Frame + glass assembly:

$$3x_1 + 2x_2 \leq 18$$

4. (Non-negativity: $x_1 \geq 0$ and $x_2 \geq 0$)

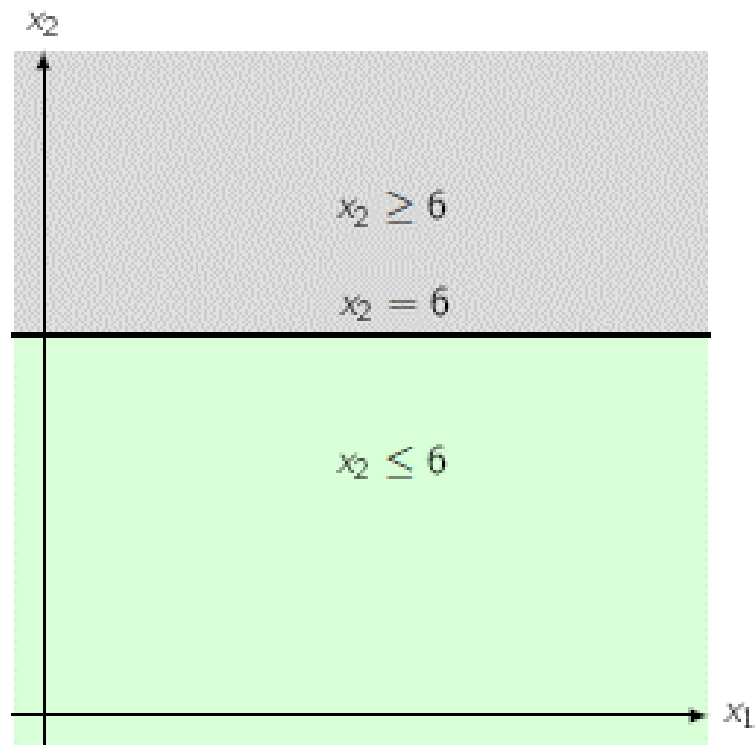
Let's have a look at the feasible region

What does the constraint $3x_1 + 2x_2 \leq 18$ do?



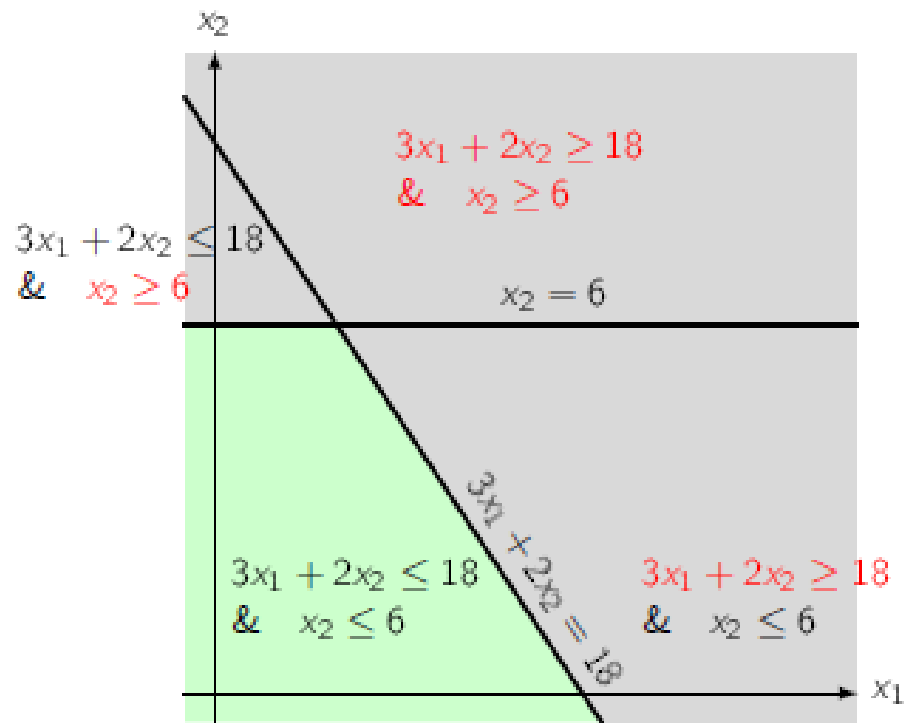
Let's have a look at the feasible region

What about $2x_2 \leq 12$?



Let's have a look at the feasible region

What does the constraint $3x_1 + 2x_2 \leq 18$ and $2x_2 \leq 12$ simultaneously



Solving the door producing problem

Door producing problem

maximize $z = 3x_1 + 5x_2$

subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

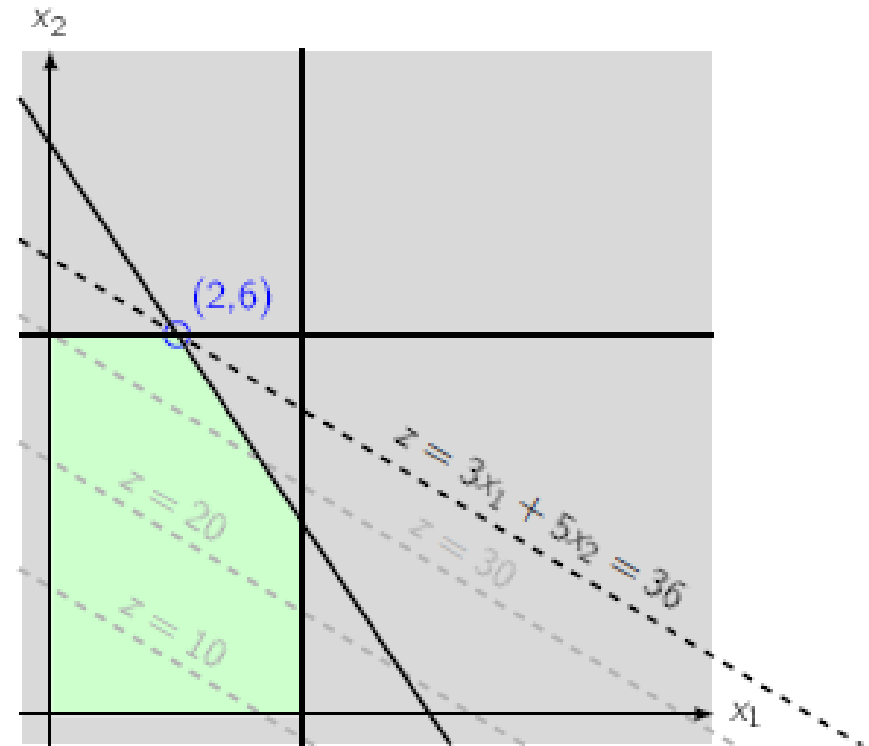
and

$$x_1, x_2 \geq 0$$

Optimal solution

$$x_1^* = 2$$

$$x_2^* = 6$$



The optimal solution of a linear program

- This example illustrates an important property with linear programs, namely

Fundamental theorem of linear programming

If an optimal solution of a linear program exists then there exists an optimal solution in the corner of the feasible region

- This holds for higher dimensions as well
- Caution, there might be **no solution** to a linear program (infeasible or unbounded).
- There might be infinitely many optimal solutions.

Infinitely many solutions

Door producing problem

maximize $z = 6x_1 + 4x_2$

subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

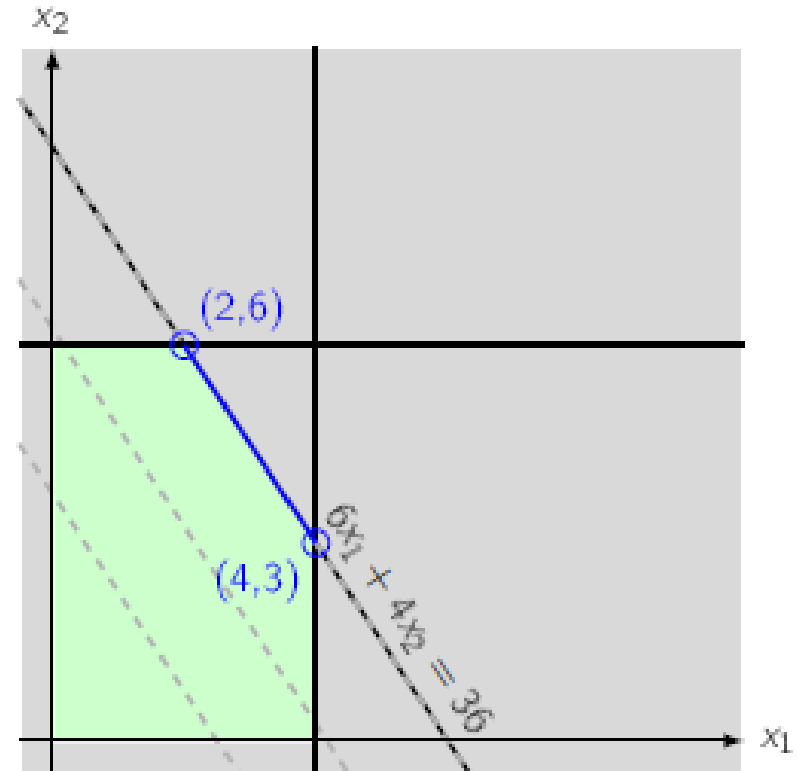
and

$$x_1, x_2 \geq 0$$

Optimal solution

$$x_1^* = 2$$

$$x_2^* = 6$$



Dual Value (Shadow Price) of a Constraint

Primal problem (standard form)

$$\max_{x \in \mathbb{R}^n} \mathbf{c}^T \mathbf{x}$$

$$\text{s.t.} \quad \mathbf{Ax} \leq \mathbf{b} \quad (\lambda)$$

$$\mathbf{x} \geq 0.$$

Where:

$$\mathbf{x} = (x_1, \dots, x_n)$$

$$\mathbf{c} = (c_1, \dots, c_n)$$

$$\mathbf{b} = (b_1, \dots, b_m)^T$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

- Each constraint $A_i x \leq b_i$ has an associated **dual variable** $\lambda_i \geq 0$
- λ_i is called the **dual value** or **shadow price** of constraint i

The dual problem

- Duality in linear programming is a very useful concept.
- Switching rules for primal and dual variables

Primal		Dual	
Objective	$Max\ z$	$Min\ z$	Objective
Variable (n)	≥ 0	≥ 0	Constraints (n)
	≤ 0	≤ 0	
	Unlimited	$=0$	
Constraints (m)	≤ 0	≥ 0	Variable (m)
	≥ 0	≤ 0	
	$=0$	Unlimited	
Right-side vector of constraints (b)		Coefficient vector of variables in objective function	
Coefficient vector of variables in objective function (c)		Right-side vector of constraints	

The dual problem

Consider the primal problem.

Primal Problem

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \quad (\lambda) \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

Then the dual problem is

Dual Problem

$$\begin{aligned} \min_{\lambda \in \mathbb{R}^m} \quad & \sum_{i=1}^m b_i \lambda_i \\ \text{s.t.} \quad & \sum_{i=1}^m a_{ji} \lambda_i \geq c_j, \quad j = 1, \dots, n \quad (x) \\ & \lambda_i \geq 0, \quad i = 1, \dots, m. \end{aligned}$$

The dual problem, Matrix notation

Consider the primal problem.

Primal Problem

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b} \quad (\lambda) \\ & \mathbf{X} \geq 0. \end{aligned}$$

Then the dual problem is

Dual Problem

$$\begin{aligned} \min_{\lambda \in \mathbb{R}^m} \quad & \mathbf{b}^T \lambda \\ \text{s.t.} \quad & \mathbf{A}^T \lambda \geq \mathbf{c} \quad (\mathbf{x}) \\ & \lambda \geq 0. \end{aligned}$$

Yes, if you are wondering, the dual problem to the dual problem is the primal problem.

Weak duality

The weak duality theorem states that

- If $\mathbf{x} \geq \mathbf{0}$ satisfies $\mathbf{Ax} \leq \mathbf{b}$
- and $\boldsymbol{\lambda} \geq \mathbf{0}$ satisfies $\mathbf{A}^T \boldsymbol{\lambda} \geq \mathbf{c}$
- then $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \boldsymbol{\lambda}$

Hence, any feasible variable of the dual problem provides an upper bound for the primal problem (and vice versa). Why?

Proof

Given feasible $\mathbf{x}$ and $\boldsymbol{\lambda}$

$$\mathbf{c}^T \mathbf{x} \leq (\mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{x} = \boldsymbol{\lambda}^T \mathbf{Ax} \leq \boldsymbol{\lambda}^T \mathbf{b} = \mathbf{b}^T \boldsymbol{\lambda}$$

Strong duality

The strong duality theorem states that

- If there exists an optimal solution x^* to the primal problem
- then there exists an optimal solution, λ^* , to the dual problem
- such that $c^T x^* = b^T \lambda^*$

Door producing company's problem revisited

Original problem

maximize $z = 3x_1 + 5x_2$

subject to:

$$x_1 \leq 4 \quad (\lambda_1)$$

$$2x_2 \leq 12 \quad (\lambda_2)$$

$$3x_1 + 2x_2 \leq 18 \quad (\lambda_3)$$

and

$$x_1, x_2 \geq 0$$

Optimal solution

$$x_1^* = 2$$

$$x_2^* = 6$$

Dual problem

minimize $z = 4\lambda_1 + 12\lambda_2 + 18\lambda_3$

subject to:

$$\lambda_1 + 3\lambda_3 \geq 3$$

$$2\lambda_2 + 2\lambda_3 \geq 5$$

and

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

Optimal solution

$$\lambda_1^* = 0$$

$$\lambda_2^* = 1.5$$

$$\lambda_3^* = 1$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 2 \end{bmatrix}$$

Economic interpretation of dual variables shadow prices

Why bother looking at the dual problem?

- Optimality certificate
 - Say I you give a linear program and you give me x^* and λ^* in return and assert that these are primal and dual optimal solutions. I can simply check that they are feasible and satisfy the strong duality criterion to see if you're right.
- Can be used in solution procedures
- If the objective has an economic meaning **the optimal dual variables can be seen as resource prices.**
 - What resources? The right hand side of the constraints, the b_i s ($i = 1, \dots, m$)
- In order to justify this interpretation we use a fact about LPs that we don't prove here: If one makes small changes to the right hand side of the primal problem then the optimal dual variables don't change.

Economic interpretation of dual variables shadow prices

Original Problem

$$\begin{aligned} \max_{x \in \mathbb{R}} \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \quad (\lambda) \\ & x \geq 0 \end{aligned}$$

Solution z^*, x^* and dual λ^*

Perturbed problem

$$\begin{aligned} \max_{x \in \mathbb{R}} \quad & z_{\Delta} = c^T x \\ \text{s.t.} \quad & Ax \leq b + \Delta b \quad (\lambda_{\Delta}) \\ & x \geq 0 \end{aligned}$$

Solution $z_{\Delta}^*, x_{\Delta}^*$ and dual λ_{Δ}^*

Provided that the elements of b are small¹, we have $\lambda^* = \lambda_{\Delta}^*$ and strong duality gives us

$$z_{\Delta}^* = (b + \Delta b)^T \lambda_{\Delta}^* = b^T \lambda_{\Delta}^* + \Delta b^T \lambda_{\Delta}^* = z^* + \Delta b^T \lambda^* = z^* + \Delta z^*$$

Thus, λ_i^* can be interpreted as the marginal cost of changing b_i .

¹ and doesn't change the basis, see Vanderbei (2008)

Door producing company's problem revisited

Original problem

maximize $z = 3x_1 + 5x_2$

subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 19$$

and

$$x_1, x_2 \geq 0$$

Optimal solution

$$x_1^* = 7/3$$

$$x_2^* = 6$$

Dual problem

minimize $z = 4\lambda_1 + 12\lambda_2 + 19\lambda_3$

subject to:

$$\lambda_1 + 3\lambda_2 \geq 3$$

$$2\lambda_2 + 2\lambda_3 \geq 5$$

and

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

Optimal solution

$$\lambda_1^* = 0$$

$$\lambda_2^* = 1.5$$

$$\lambda_3^* = 1$$

The new optimal objective is $z^* = 37$, i.e. the change is $z_{\Delta}^* = \lambda_3^* \Delta b_3$

What is the price of a non-binding constraint?

- How much are you willing to pay for something you already more than enough of?
 - Nothing right?
- For all the less than or equal constraints $\sum_{j=1}^n a_{ij}x_j \leq b_i$ we have

$$\left(\sum_{j=1}^n a_{ij}x_j^* - b_i \right) \lambda_i^* = 0 \quad i = 1, \dots, m,$$

for optimal x^* and corresponding (optimal) dual λ^*

- λ^* can therefore only be non-zero if constraint i is binding
- Called the complementary slackness condition.
- If λ_i^* is interpreted as the price you are willing to pay for a resource b_i then the it will be zero unless having more (or less) of the resource improves (or worsen) your objective.

Door producing company's problem revisited

Original problem

maximize $z = 3x_1 + 5x_2$

subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1, x_2 \geq 0$$

Optimal solution

$$x_1^* = 2$$

$$x_2^* = 6$$

Dual problem

minimize $z = 4\lambda_1 + 12\lambda_2 + 18\lambda_3$

subject to:

$$\lambda_1 + 3\lambda_2 \geq 3$$

$$2\lambda_2 + 2\lambda_3 \geq 5$$

and

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

Optimal solution

$$\lambda_1^* = 0$$

$$\lambda_2^* = 1.5$$

$$\lambda_3^* = 1$$

First constraint non-binding so $\lambda_1^* = 0$

Summary

You have now seen

- An example of a linear program
- That, indeed, in this linear program the optimal solution was in the corner of the feasible region
 - (and of course you believe me when I tell you that this is in general the case, also in higher dimensions)
- That for a linear program there is an associated problem. called the dual problem and that the two problems are linked.
- The optimal solution to the dual program can often be regarded as resource prices.
- That the resource prices are non-zero only if the resource is fully used in the optimal solution (biding constraint).